

Solução do Problema Bore3d - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: Bore3d
- Número de Variáveis: 315
- Número de Restrições: 233
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 0.000e+00
- Valor Primal: 1.373e+03
- Valor Dual: 0.000e+00
- Gap: 0.000e+00
- Número de Iterações: 13

2 Variáveis Primais e Custos Reduzidos ($x \geq 0$ e $z \geq 0$)

Tabela 1: Variáveis primais (x) e custos reduzidos (z) com valores ≥ 0 do problema Bore3d

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	2.933e+00	2	3.580e+00
18	8.372e-01	3	7.967e+01
21	3.039e+00	4	2.933e+00
55	3.206e+00	5	1.712e+01
57	3.970e+00	6	5.235e+01
61	1.961e+00	7	4.710e+01
62	5.208e-01	8	4.575e+01
63	3.717e-01	9	1.438e+01
64	1.215e-01	11	6.776e+00
65	7.978e-01	12	1.565e+02
66	1.795e+00	14	6.611e+01
67	1.793e+01	15	7.420e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
68	5.677e+01	16	7.420e-01
74	1.934e+01	17	7.420e-01
75	1.020e+01	19	4.623e-01
80	5.034e+02	22	9.623e-01
81	5.357e-01	23	1.270e+00
84	4.402e-01	24	9.623e-01
85	3.194e-01	26	8.494e-01
87	2.879e-01	28	4.494e-01
88	9.859e-01	29	8.494e-01
91	3.234e+01	30	1.270e+00
93	4.747e+00	31	9.623e-01
96	1.030e+01	32	1.270e+00
97	8.458e-01	33	9.623e-01
106	7.922e+00	35	8.494e-01
107	5.997e+00	37	5.623e-01
108	6.209e+00	38	4.494e-01
109	2.308e+00	39	8.494e-01
110	9.396e+03	40	1.270e+00
111	9.315e-01	42	8.494e-01
112	9.315e-01	43	9.623e-01
114	1.215e-01	44	8.494e-01
117	1.563e+01	45	8.494e-01
118	1.282e+00	46	9.623e-01
119	2.546e+00	47	1.270e+00
121	2.325e+00	48	9.623e-01
124	1.563e+01	50	8.494e-01
125	6.732e+00	52	4.494e-01
126	2.324e+01	53	8.494e-01
127	7.692e+00	54	1.270e+00
128	7.692e+00	67	7.657e+01
129	3.032e+00	69	3.343e+00
130	2.970e+00	70	2.487e+01
131	1.973e+01	71	7.780e-01
132	7.978e-01	72	2.643e+00
133	1.212e-01	76	4.726e+00
134	1.212e-01	77	8.686e+01
135	4.273e+00	78	3.575e+00
136	4.273e+00	79	1.072e+01
137	4.521e+00	82	1.781e+00
138	4.922e+01	83	1.011e+00
139	2.115e+00	92	2.058e+00
140	1.533e+00	94	2.754e+00
141	1.876e+01	95	1.546e+00
142	1.876e+01	98	3.245e+01
143	1.876e+01	99	1.382e+01
146	2.679e-02	101	1.664e+00
147	2.679e-02	102	1.959e+01
148	5.117e+00	103	1.996e-01
149	2.325e+00	104	1.467e+02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
151	1.241e+03	105	1.194e+00
152	1.000e+01	113	4.258e+00
153	1.563e+01	115	7.643e+00
155	1.201e+00	116	6.199e+00
162	1.972e+00	120	3.695e+00
169	7.708e-01	122	8.898e+00
171	1.533e+00	123	2.724e+00
172	4.402e-01	144	2.028e-01
173	3.194e-01	145	2.107e+01
180	7.708e-01	156	3.533e+01
194	5.357e-01	158	3.137e+01
196	4.747e+00	159	4.728e+01
197	1.030e+01	160	2.371e+01
198	8.458e-01	161	2.371e+01
211	1.795e+00	163	2.371e+01
214	2.298e+00	164	2.837e+01
215	2.115e+00	166	2.014e+01
217	1.154e+00	167	1.350e+02
218	5.417e-01	170	1.649e+00
219	3.738e-01	174	6.968e+00
220	2.381e-01	175	5.111e+00
221	9.315e-01	177	1.019e+01
222	1.563e+01	178	1.051e+01
224	4.655e-01	181	1.743e+01
227	8.161e-01	182	1.743e+01
230	2.546e+00	183	1.075e+01
231	2.325e+00	184	5.955e+00
232	1.563e+01	185	7.078e+00
233	6.732e+00	187	3.297e+01
234	7.692e+00	188	3.034e+01
242	2.748e+00	189	2.527e+01
245	1.795e+00	190	2.689e+01
246	1.793e+01	191	4.414e+00
247	1.212e-01	192	2.715e+01
248	4.273e+00	193	5.110e+00
252	2.115e+00	195	2.620e+00
255	1.876e+01	199	3.354e+02
256	2.679e-02	200	3.354e+02
257	5.117e+00	201	3.096e+02
258	1.937e+00	202	3.354e+02
259	7.692e+00	203	3.354e+02
261	8.542e+03	204	3.001e+02
263	5.940e-01	205	3.004e+02
264	3.777e+01	206	3.354e+02
266	1.201e+00	207	1.743e+01
267	7.708e-01	208	1.743e+01
268	5.677e+01	209	3.412e+00
269	1.759e+01	210	3.319e+00
270	4.577e+02	213	4.562e+01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
271	2.113e+01	216	3.251e+01
274	4.696e+03	229	6.463e+00
275	4.064e+01	235	2.542e+01
277	4.042e+00	236	2.609e+01
278	2.146e+00	238	2.560e+01
280	7.226e+00	239	1.083e+01
281	3.980e+02	240	1.382e+01
282	8.178e+00	241	3.504e+01
283	7.428e+03	244	3.534e+01
285	1.345e+01	249	9.588e+00
286	5.166e-01	250	4.837e+01
287	5.296e+03	251	8.781e+00
296	1.129e+03	253	4.168e+00
302	3.650e+00	254	3.136e+00
303	2.933e+00	260	1.017e-01
308	3.968e+00	262	9.583e-02
309	3.876e+00	265	2.660e-01
		272	3.677e-01
		273	1.702e-01
		276	1.455e-01
		284	1.957e-01
		288	2.909e+02
		289	2.909e+02
		290	3.353e+02
		291	3.353e+02
		292	3.354e+02
		293	2.099e-03
		294	2.841e-03
		295	4.440e-03
		297	9.199e-03
		298	2.357e-03
		299	3.577e-03
		300	4.627e-03

3 Variáveis Duais (Multiplicadores de Lagrange - Valores λ 0)

Tabela 2: Variáveis duais (y) com valores λ 0 do problema Bore3d

Coordenada y	Valor y (Dual)
2	5.376e-01
21	2.323e-01
23	2.038e-01
24	1.589e-01
27	1.000e+00
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Coordenada y	Valor y (Dual)
30	1.000e+00
34	1.000e+00
35	1.000e+00
36	1.000e+00
37	1.000e+00
42	1.022e-01
44	1.499e-01
54	1.000e+00
55	1.000e+00
56	1.000e+00
61	3.509e-01
63	8.422e+00
65	4.799e-01
68	4.104e+00
81	3.343e+01
82	6.269e+01
85	1.000e+01
90	1.000e+00
93	1.000e+00
94	1.000e+00
95	1.000e+00
96	1.000e+00
97	2.300e+00
103	8.031e-02
104	1.260e+00
109	1.064e-02
110	1.133e+01
113	3.583e+01
114	8.101e+00
115	3.691e+00
116	2.611e+00
117	2.241e+00
118	5.242e+01
119	3.799e+00
120	2.471e-01
122	2.282e+00
125	5.311e-01
126	5.274e-01
127	4.054e+01
130	3.354e+02
136	4.855e-01
137	2.997e+01
138	1.830e-01
139	8.568e+00
140	3.030e-03
141	1.864e+00
142	6.401e-01
143	6.878e-01
145	3.670e+01
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Coordenada y	Valor y (Dual)
148	4.546e+00
149	1.035e+01
150	1.875e+00
152	3.839e+00
155	1.298e+00
156	1.252e+00
157	5.526e+00
158	1.276e-01
159	4.180e-01
160	1.454e-01
161	9.677e-01
162	7.028e+00
163	7.797e-01
164	5.201e-01
165	2.659e+01
166	6.720e+00
167	1.740e+00
168	1.219e+00
169	7.410e-03
170	6.044e-01
171	1.267e-01
172	7.221e-02
173	1.550e+00
174	2.743e+00
175	9.070e-03
176	1.624e+00
177	2.764e-01
180	1.585e+02
181	5.206e+01
182	2.131e+00
183	2.195e-01
184	3.769e-01
185	4.080e-03
186	3.100e-04
187	9.444e-01
190	3.535e-01
191	5.122e-02
192	3.711e-01
195	1.573e+01
196	3.833e-03
197	2.300e+00
201	3.074e-01
202	4.454e-02
203	2.855e-01
208	3.485e-03
209	1.163e-01
212	2.024e-01
213	3.126e-01
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Coordenada y	Valor y (Dual)
214	3.833e-03